What Does a Machine Know When It Checks a Proof?

Philosophical Implications of the Cathedral Formalization

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Abstract

The *Cathedral* is a machine-verified formalization in the Lean 4 theorem prover establishing that the Riemann Hypothesis is logically equivalent to the decay of the Nyman–Beurling distance. It does not prove the Riemann Hypothesis. It proves, with certainty beyond human capacity for error, exactly what one *would* need to prove. This paper examines the philosophical questions raised by this artifact: What is the epistemic status of a conditional proof? What does it mean for a machine to "catch" a mathematical error that a human would not? Can axioms serve as units of mathematical knowledge, decomposing an open problem into precisely measured fragments of remaining ignorance? We argue that the Cathedral represents a new category of mathematical object—the *certified reduction*—that is neither a proof nor a conjecture, but a machine-witnessed guarantee about the logical structure of an unsolved problem.

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1 Introduction: A New Kind of Mathematical Object

On April 19, 2026, a 78-file computer program was compiled without error. The program, written in the Lean 4 proof language and verified against the Mathlib mathematical library, established the following:

The Riemann Hypothesis is true if and only if $d_N^2 \to 0$*,*

where d_N^2 is a concrete quantity measuring how well certain sawtooth functions approximate a constant. The biconditional is complete: both directions are proved. The proof depends on exactly 2 mathematical axioms (statements accepted without proof) plus 3 logical axioms built into Lean's type theory.

The Riemann Hypothesis remains unproved. But something has changed. Before the Cathedral, the logical structure of the problem was distributed across hundreds of papers, written over 167 years, in inconsistent notation, with varying standards of rigor. Now it exists as a single, machine-auditable artifact in which every deductive step has been verified, every assumption is named, and the frontier of human ignorance is precisely demarcated.

This artifact raises questions that mathematics alone cannot answer. They are questions for philosophy.

2 The Epistemic Status of Conditional Proof

2.1 What Has Been Proved?

The Cathedral proves a biconditional: RH $\iff$ $d_N^2 \to 0$. It does *not* prove either side independently. Neither the truth nor the falsity of RH follows from the Cathedral alone.

This is not a new situation in mathematics. Many theorems are conditional: "If the Generalized Riemann Hypothesis is true, then. . . " What *is* new is the machine verification. The biconditional is not a claim in a paper that might contain an error. It is a statement that has been checked by a computer against a formal system of logic (the Calculus of Inductive Constructions), in which every step follows from the axioms by mechanical rule application.

> *The conditional is certain. The condition is not.*

This creates an unusual epistemic asymmetry. We know, with a certainty that exceeds any human proof, that RH is equivalent to $d_N^2 \to 0$. But we do not know whether either statement is true. The certainty attaches to the *structure* of the implication, not to its content.

2.2 Is a Biconditional a "Proof"?

In the formalist tradition (Hilbert, Bourbaki), a proof is a finite sequence of deductions from axioms. By this standard, the Cathedral *is* a proof—specifically, it is a proof of the proposition "RH $\iff$ $d_N^2 \to 0$." The proposition happens to be a biconditional rather than a simple statement, but formalism makes no distinction between the two.

In the intuitionist tradition (Brouwer, Heyting), a proof of $P \iff Q$ is a pair of constructions: one transforming a proof of P into a proof of Q, and vice versa. The Cathedral provides exactly this: the Lean file `Assembly/MainChain.lean` contains a term of type RH $\to$ $(d_N^2 \to 0)$ and a term of type $(d_N^2 \to 0) \to$ RH. (Lean 4 uses classical logic via `Classical.choice`, so the constructions are not fully constructive, but they are syntactically present.)

In the Lakatosian tradition (*Proofs and Refutations*), a proof is a social process of conjecture, criticism, and revision. By this standard, the Cathedral is profoundly proof-like: it was built through 23 days of iterative refinement, with 96 archived files documenting failed approaches, false lemmas, and superseded strategies. The machine played the role of Lakatos's "critic," catching errors that the human would have missed (Section 4).

2.3 The Axiom as a Unit of Ignorance

The most philosophically interesting feature of the Cathedral is its treatment of axioms. In classical foundations (ZFC), axioms are the starting points of all mathematics—they are accepted because they are self-evident or because they codify our best intuitions about sets.

In the Cathedral, axioms serve a fundamentally different purpose. They are *placeholders for unformalized knowledge.* Each axiom represents a specific, named, precisely stated piece of mathematics that is believed to be true (and has been proved by humans in traditional papers) but has not yet been translated into Lean.

This makes the axiom a unit of *measured ignorance.* The Cathedral does not merely say "we don't know how to prove RH." It says: "We don't know how to prove RH, and here are the *exactly two* facts we would need to formalize in order to reduce it to a question about L^2 approximation."

An axiom is not a failure of proof. It is a unit of measurement for what remains unproved.

The reduction from 56 axioms (March 27, Day 1) to 2 axioms (April 18, Day 22) is not merely a technical achievement. It is an epistemic one: the frontier of ignorance was compressed by 96% in three weeks.

3 What Does the Machine "Know"?

3.1 Machine Verification Is Not Understanding

The Lean type-checker that verifies the Cathedral does not "understand" the Riemann Hypothesis. It does not know what a prime number is, what the zeta function does, or why the problem matters. It checks that each step in the proof follows from the previous one by the rules of the Calculus of Inductive Constructions. It performs this check with perfect reliability, but without any semantic comprehension.

This is a philosophical distinction with practical consequences. A human mathematician who understands a proof can:

- Generalize it to new contexts.
- Identify which steps are "deep" and which are routine.
- Predict where the proof technique might fail.
- Extract intuition for new conjectures.

The Lean type-checker can do none of these things. It can verify but not understand, certify but not explain.

3.2 But Verification Has Its Own Epistemic Value

The flip side is that the machine verification provides something that human understanding cannot: *certainty.* Human proofs contain errors. The history of mathematics is littered with proofs that were accepted for years or decades before flaws were discovered (Kempe's proof of the Four-Color Theorem, Wiles's initial proof of Fermat's Last Theorem, Voevodsky's motivating experience with homotopy groups).

The Cathedral's verification eliminates this class of error entirely. If the Lean kernel is correct (a much smaller and more auditable artifact than any mathematical proof), then the biconditional holds. Period.

This creates a division of epistemic labor:

Machine provides	Human provides
Deductive certainty	Semantic understanding
Error elimination	Intuition and creativity
Axiom tracking	Mathematical strategy
Exhaustive checking	Selective attention

Neither alone is sufficient. Together, they produce an artifact with properties that neither could achieve independently.

3.3 The AI Collaborator as Philosophical Agent

The Cathedral was built not by a human alone, nor by a machine alone, but by a human-AI pair. The AI assistant (an LLM-based coding agent) wrote most of the Lean code, discovered Mathlib API functions, and proposed proof strategies. The human directed the mathematical architecture and made all strategic decisions.

This raises a question: *who* proved the biconditional?

- The human, who directed the strategy? But the human could not have written the 20,000 lines of Lean code alone.
- The AI, who wrote the code? But the AI did not understand the mathematics and could not have chosen the proof architecture.
- The Lean kernel, which verified the proof? But the kernel made no creative contribution—it only checked.

Perhaps the correct answer is that mathematical proof is becoming a *collective* activity in a new sense: not a team of humans, but a team of heterogeneous cognitive agents, each contributing a different epistemic capacity. The result is a proof that no single agent could have produced.

4 Five Traps and the Limits of Informal Reasoning

During construction, the machine-verification process detected five mathematical errors that would have been invisible to informal proof. Each illuminates a different limitation of human mathematical reasoning:

4.1 The Basis Trap

Two natural choices for the sawtooth basis—$\{k/x\}$ and $\{1/(kx)\}$—appear interchangeable to informal reasoning. They are not. The first generates $L^2(0,1)$ unconditionally (independent of RH), making any approximation result trivially true. The second has approximation quality governed by the zeta zeros.

An informal proof using the wrong basis would appear correct at every step. No referee would catch it, because the error is in the *choice of definition*, not in any deductive step. The machine caught it because the Lean type system enforces that the basis functions must match the type signature of the Nyman–Beurling criterion.

Philosophical lesson: Definitions are not epistemically neutral. The choice of how to formalize a concept constrains which theorems can be proved about it. Machines enforce definitional consistency in a way that human attention cannot.

4.2 The Cancellation Trap

The triangle inequality gives $\|1-f\|_2 \leq 1+\|f\|_2$, yielding $d_N^2 \leq 4$ for a quantity approaching 0. The bound is mathematically correct but useless, because it destroys the cancellation $1 - 2b^T v + v^T G v \to 0$ that makes d_N^2 small.

An informal proof might invoke the triangle inequality without noticing that it loses the essential structure. The machine caught this because the Lean proof could not close the gap between the bound and the target.

Philosophical lesson: Not all valid inequalities are useful. Some destroy the very structure they are meant to control. This is invisible to syntactic reasoning but immediately apparent to a type-checker that demands a complete chain from hypothesis to conclusion.

4.3 The Ghost Axiom Trap

Nine mathematical propositions existed simultaneously as axioms (unproved) and theorems (proved) in different files. The module system allowed both to coexist without contradiction. Downstream theorems could silently depend on the axiom rather than the theorem, inflating the proof's apparent weakness.

Philosophical lesson: In modular epistemic systems, the same proposition can have different justification statuses in different contexts. Coherence checking across modules is a non-trivial epistemic task that humans perform poorly and machines perform well.

5 The Cathedral as Philosophical Object

5.1 A Certified Reduction

We propose that the Cathedral represents a new category of mathematical object: the *certified reduction*. A certified reduction has three components:

1. A **target proposition** (here: RH).
2. A **reduced proposition** (here: $d_N^2 \to 0$).
3. A **machine-verified biconditional** connecting them, with every assumption explicitly named.

A certified reduction is stronger than a conjecture (it establishes logical equivalence) but weaker than a proof (it does not resolve either proposition). Its epistemic value lies in the precision of the residual: the named axioms constitute an *exact measure* of what remains unknown.

5.2 Axiom Reduction as Epistemic Progress

In traditional mathematics, progress on an open problem is measured by theorems: new results that narrow the gap between what is known and what is conjectured. The Cathedral suggests a complementary measure: *axiom count on the critical path.*

Axioms	Interpretation
56	"We have a rough sketch and many assumptions."
12	"We have a solid architecture with specific gaps."
5	"We can name every remaining piece."
2	"Two facts stand between us and certified equivalence."
0	"The equivalence is fully machine-verified."

The progression from 56 to 2 is epistemic progress, even though neither RH nor its negation was established. The frontier of ignorance shrank. The remaining gaps were named, located, and measured. This is a form of understanding that traditional mathematics—which classifies results as "proved" or "open" with no gradations—does not capture.

5.3 The Archaeology of Failed Proofs

The Cathedral's archive of 96 superseded files is, to our knowledge, unprecedented in mathematical practice. Mathematicians typically discard failed approaches. The Cathedral preserves them, creating a *fossil record* of mathematical exploration.

This archive has philosophical significance. It makes visible the *negative knowledge* that is normally invisible: approaches that don't work, lemmas that are false, definitions that lead nowhere. Lakatos argued that refutations are as important as proofs. The Cathedral's archive is a concrete embodiment of this principle.

The 54% discard rate (96 archived files out of 174 total) quantifies something that mathematics has never measured before: the ratio of exploration to exploitation in a proof search. More than half of all mathematical effort was invested in paths that were eventually abandoned. This is not failure; it is the cost of discovery.

6 Implications for Mathematical Practice

6.1 The End of the Informal Era?

The Cathedral was not built because informal mathematics is inadequate for the Nyman–Beurling criterion. Every theorem in the Cathedral could, in principle, be written as a traditional paper proof. The machine verification adds no new mathematical content.

What it adds is *trust infrastructure.* In a world where LLMs can generate plausible-sounding but incorrect mathematical proofs, machine verification provides a trust anchor that is independent of human judgment, reputation, or social consensus.

> *Lean does not care who you are. It cares whether your proof type-checks.*

This is both a democratization (anyone can submit a machine-verified proof) and a raising of standards (no amount of prestige can override a type error).

6.2 The Decomposition of Mathematical Problems

If axioms can serve as units of ignorance, then open problems can be *decomposed* into precisely stated sub-problems, each corresponding to an axiom. This transforms the sociology of mathematical research: instead of one person (or group) attempting an entire proof, the problem can be distributed.

The Cathedral's 2 remaining axioms are, in effect, *bounty postings*: precisely stated mathematical challenges, with a machine-verified guarantee that formalizing them would complete the reduction. This is a new mode of mathematical collaboration, enabled by the precision of formal systems.

6.3 What Counts as a Contribution?

Traditional mathematics values theorems. A mathematician's contribution is measured by the propositions they prove. The Cathedral suggests a broader accounting:

- **Axiom elimination** (proving a previously assumed result) is a contribution.
- **Architecture design** (choosing the right decomposition of a problem into modules) is a contribution.
- **Bug detection** (finding the basis trap, the cancellation trap) is a contribution.
- **Archival** (documenting failed approaches so others don't repeat them) is a contribution.

None of these fit neatly into the "theorem proved" metric. But all of them advanced mathematical knowledge in a measurable way.

7 The Question of Beauty

Mathematicians often speak of beauty in proofs. Hardy's *A Mathematician's Apology* elevates aesthetic criteria to central importance. The Cathedral raises the question: can a machine-verified proof be beautiful?

The Rank-1 Mellin Miracle is, by any standard, an elegant result. At a zeta zero ρ, the Mellin transform of the Báez-Duarte basis function collapses to $\mathcal{M}[h_k](\rho) = 1/(k(\rho - 1))$.

The entire infinite-dimensional computation reduces to a single multiplication. The k-dependence and ρ-dependence factorize completely.

This is beautiful. And it was discovered during machine verification, when a general strategy (Hahn–Banach separation) was replaced by a specific computation that turned out to have unexpected structure. The machine did not appreciate the beauty. The human did. But the beauty would not have been found without the machine's insistence on explicit computation.

> *The machine demands rigor. The human supplies meaning. Neither alone produces beauty. Together, sometimes, they do.*

8 Conclusion: The Cathedral as Epistemological Experiment

The Cathedral is many things: a 78-file Lean program, a reduction of the Riemann Hypothesis, a case study in proof engineering, a record of human-AI collaboration. But above all, it is an *epistemological experiment*: an attempt to make the structure of mathematical knowledge visible, precise, and machine-auditable.

The experiment yields several findings:

1. **Conditional proofs have precise epistemic value**, measurable by the number and nature of their axioms.
2. **Machine verification and human understanding are complementary**, not competitive, epistemic capacities.
3. **Axioms can serve as units of measured ignorance**, decomposing open problems into named, located, precisely stated fragments.
4. **Failed approaches have epistemic value** when archived systematically.
5. **Mathematical beauty can emerge from machine-assisted exploration**, though it requires human recognition to be appreciated.

Two axioms remain. The Riemann Hypothesis is neither proved nor disproved. But the space of possible proofs has been mapped with unprecedented precision, and that mapping is itself a form of mathematical knowledge—one that our philosophical vocabulary is only beginning to accommodate.

The Cathedral does not answer the question.
It makes the question precise.
That is philosophy's oldest aspiration.

Source code: `https://github.com/jrgochan/prime`

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